

# RaceDay — API Endpoint Plan (Part 1, Section B)

## Authentication Endpoints

| HTTP Method | Route              | Description             | Role Required | Request Body                | Expected Response                |
|-------------|--------------------|-------------------------|---------------|-----------------------------|----------------------------------|
| POST        | /api/auth/register | Register a new user     | None          | Name, Email, Password, Role | 201 Created, 400 Bad Request     |
| POST        | /api/auth/login    | Authenticate user       | None          | Email, Password             | 200 OK (Token), 401 Unauthorized |
| POST        | /api/auth/logout   | End the current session | Any           | None                        | 200 OK                           |

## Profile Endpoints

| HTTP Method | Route        | Description        | Role Required | Request Body       | Expected Response       |
|-------------|--------------|--------------------|---------------|--------------------|-------------------------|
| GET         | /api/profile | View own profile   | Any           | None               | 200 OK                  |
| PUT         | /api/profile | Update own profile | Any           | Name, Email, Phone | 200 OK, 400 Bad Request |

## Event Endpoints

| HTTP Method | Route            | Description             | Role Required | Request Body          | Expected Response     |
|-------------|------------------|-------------------------|---------------|-----------------------|-----------------------|
| GET         | /api/events      | Retrieve all events     | Any           | None                  | 200 OK                |
| GET         | /api/events/{id} | Retrieve specific event | Any           | None                  | 200 OK, 404 Not Found |
| POST        | /api/events      | Create new event        | Organiser     | Event details         | 201 Created           |
| PUT         | /api/events/{id} | Update existing event   | Organiser     | Updated event details | 200 OK, 404 Not Found |
| DELETE      | /api/events/{id} | Delete event            | Organiser     | None                  | 200 OK, 404 Not Found |

## Category Endpoints

| HTTP Method | Route                       | Description                      | Role Required | Request Body             | Expected Response     |
|-------------|-----------------------------|----------------------------------|---------------|--------------------------|-----------------------|
| GET         | /api/events/{id}/categories | Retrieve categories for an event | Any           | None                     | 200 OK                |
| POST        | /api/events/{id}/categories | Add a category to an event       | Organiser     | Category details         | 201 Created           |
| PUT         | /api/categories/{id}        | Update a category                | Organiser     | Updated category details | 200 OK, 404 Not Found |
| DELETE      | /api/categories/{id}        | Delete a category                | Organiser     | None                     | 200 OK, 404 Not Found |

## Enrolment Endpoints

| HTTP Method | Route                           | Description                      | Role Required | Request Body | Expected Response         |
|-------------|---------------------------------|----------------------------------|---------------|--------------|---------------------------|
| POST        | /api/categories/{id}/enrolments | Enrol in a category              | Participant   | None         | 201 Created, 409 Conflict |
| GET         | /api/enrolments/my              | View own enrolments              | Participant   | None         | 200 OK                    |
| GET         | /api/events/{id}/enrolments     | View all enrolments for an event | Organiser     | None         | 200 OK, 404 Not Found     |

## Result Endpoints

| HTTP Method | Route             | Description          | Role Required | Request Body                      | Expected Response     |
|-------------|-------------------|----------------------|---------------|-----------------------------------|-----------------------|
| GET         | /api/results/my   | View own results     | Participant   | None                              | 200 OK                |
| GET         | /api/results/{id} | View specific result | Any           | None                              | 200 OK, 404 Not Found |
| POST        | /api/results      | Capture race result  | Organiser     | EnrolmentId, FinishTime, Position | 201 Created           |
| PUT         | /api/results/{id} | Update a result      | Organiser     | Updated result                    | 200 OK, 404 Not Found |

## Location & Weather Endpoints

| HTTP Method | Route                      | Description                                                                               | Role Required | Request Body | Expected Response                                                     |
|-------------|----------------------------|-------------------------------------------------------------------------------------------|---------------|--------------|-----------------------------------------------------------------------|
| GET         | /api/categories/{id}/route | Retrieve route details (distance, elevation, start/end point) for a category              | Any           | None         | 200 OK, 404 Not Found                                                 |
| GET         | /api/events/{id}/weather   | Retrieve live weather for the event's location (server-side call to external weather API) | Any           | None         | 200 OK, 404 Not Found, 502 Bad Gateway (weather provider unavailable) |

**Note on directions:** getting directions to an event does not need its own API endpoint — the MVC frontend (Part 3) builds a "Get Directions" link straight to Google Maps using the event's stored `Location`/coordinates. No map API key or backend call is required for that piece.

**Role-based access summary:**

- **Public / no login:** browse events, view categories, view specific results.
- **Any (logged in, either role):** view/edit own profile.
- **Participant only:** enrol in a category, view own enrolments, view own results.
- **Organiser only, and only for events/categories/results they own:** create/edit/delete events, manage categories, view all enrolments for their event, capture results.